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RESEARCH PAPER

MdRAD5B regulates the chromatin accessibility of *MdMYB14* under cold by associating with a transcription factor MdLSD1

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RADIATION SENSITIVE 5B (MdRAD5B) has previously been identified as a chromatin remodeler involved in regulating the chromatin accessibility and H3K27me3 modification of genes. However, how RAD5B regulates the chromatin accessibility of target genes remains largely unclear. Here, we find that MdRAD5B positively regulates apple cold tolerance by modulating the expression of cold-responsive genes. Under cold stress, MdRAD5B regulates the expression of cold-responsive genes, including *MdMYB14*, through its chromatin remodeling function. Further study reveals that the transcription factor LESION SIMULATING DISEASE 1 (MdLSD1) recruits MdRAD5B and reduces the chromatin accessibility of *MdMYB14*. MdMYB14, a downstream target of MdLSD1 and a negative regulator of cold tolerance, binds to the promoter region of *MdCBF3* and inhibits its expression. Additionally, we identified a natural variation, a non-synonymous mutation at position 1,780 of the *RAD5B* CDS (*RAD5B*^{CDS1780}), which affects apple cold tolerance by altering the ATPase activity of RAD5B. Taken together, our results unravel the underlying molecular mechanism by which MdRAD5B negatively regulates the chromatin accessibility of *MdMYB14* through MdLSD1 in apple responses to cold.

MdRAD5B | chromatin accessibility | MdLSD1 | cold tolerance

INTRODUCTION

The expression of eukaryotic genes is influenced by various factors, including *cis*-regulatory elements, *trans*-acting factors, chromatin DNA, histone modifications, and chromatin spatial structures (Kim, 2021). Although the molecular mechanism by which chromatin regulates eukaryotic gene expression through spatial structural changes is not yet fully understood, numerous epigenetic studies have revealed that the spatial structure of chromatin plays a crucial role in this process. The process of chromatin remodeling is mainly affected by ATP-dependent chromatin remodelers and histone modifiers. Among them, chromatin remodelers drive the "sliding" and "displacement" of nucleosomes mediated by chromatin complexes hydrolyzing ATP to release energy, thereby achieving chromatin remodeling and altering chromatin structure (Clapier et al., 2017; Kim, 2021).

To date, most identified chromatin remodelers belong to the SNF2 family, which alter chromatin structure to regulate gene expression in response to diverse stresses (Song et al., 2021). CHR19 is an SNF2 family protein that can influence gene transcription through its chromatin remodeling function to cope with different pathogens (Kang et al., 2022). Maize *ZmCHB101* is a homolog of *Arabidopsis SWI3D*. Under osmotic stress or salt stress, the survival rate of maize plants with suppressed *ZmCHB101* expression is significantly lower than that of wild-type (WT) plants. In addition, ZmCHB101 affects gene expression at the transcriptional level by regulating the nucleosome density of stress-responsive genes and the enrichment of RNA polymerase II at transcription start sites (Yu et al., 2018). Furthermore, in *Arabidopsis*, the BRM protein of the SWI/SNF family interacts with HD2C (HISTONE DEACETYLASE 2C) and regulates the heat stress response (Buszewicz et al., 2016). Similarly, the PICKLE protein of the CHD family forms a molecular switch for cold acclimation by maintaining an open chromatin state and cooperating with the transcription factor ICE1 to activate *CBF* genes (Yang et al., 2019). In rice, OsCHR11 promotes chromatin opening in the promoter region of genes and enhances cold tolerance (Li et al., 2024). Apple MdRAD5B, which belongs to the SNF2 protein family, improves drought tolerance by modulating the expression of drought-responsive genes through the regulation of chromatin remodeling and H3K27me3 deposition (Song et al., 2024).

Chromatin remodeling factors themselves usually do not possess sequence-specific DNA-binding abilities (Clapier et al., 2017). Therefore, exploring how chromatin remodeling factors achieve precise regulation of specific regions has become a

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hotspot in recent years (Huang et al., 2025). Recent studies have shown that their localization is mediated through interactions with other molecular entities, primarily through three mechanisms. (i) Association with sequence-specific transcription factors (TFs). For example, during thermomorphogenesis, the key transcription factor PHYTOCHROME INTERACTING FACTOR 4 (PIF4) directly recruits the chromatin remodeling INO80 complex to its target loci (Xue et al., 2021). Additionally, a plant pioneer TF, LFY (LEAFY), can recruit BRM and SYD to activate floral fate (Jin et al., 2021). (ii) Recognition of histone modification marks. In tumor cells, the SWI/SNF PBAF complex binds to H3K4me3 modifications, directing the remodeler to promoters of actively transcribed genes (Grossi et al., 2025). In plants, the recruitment of the SWR1 chromatin remodeling complex to genes marked by H3K4me3 is facilitated by AL proteins, which specifically bind to this histone modification (Su et al., 2025). (iii) Guidance by DNA methylation. Methyl-binding proteins such as SUVH9 can recruit chromatin remodelers, including DDM1, to maintain heterochromatin condensation and suppress transposon activity (Jing et al., 2016).

As a critical environmental factor, cold stress exerts profound effects on plant growth, productivity, and survival, thereby restricting the geographical distribution of numerous plant species (Ritonga and Chen, 2020). Cold stress triggers the accumulation of reactive oxygen species (ROS), such as hydrogen peroxide (H_2O_2), hydroxyl radicals, and superoxide anions (O_2^-), leading to secondary oxidative stress (Meng et al., 2025; Waszczak et al., 2018). To cope with cold stress, plants have evolved sophisticated molecular mechanisms, with CBF transcription factors playing a central role in plant responses to cold stress (Jia et al., 2016; Novillo et al., 2007; Thomashow, 1999; Wang et al., 2025; Zhao et al., 2016). Under cold conditions, *CBF* genes are induced, which then activate cold-regulated (COR) genes, thereby enhancing cold tolerance (Shi et al., 2018; Thomashow, 1999).

Previous research has shown that MdRAD5B regulates the chromatin accessibility of many genes (Song et al., 2024); however, how it regulates gene expression is largely unknown. In this study, we found that MdRAD5B facilitates the expression of cold-responsive genes, including *MdMYB14*. MdLSD1 directly binds to the *MdMYB14* promoter and recruits MdRAD5B, thereby reducing the chromatin accessibility of *MdMYB14* and inhibiting its expression under cold conditions. MdMYB14 further binds to the promoter of *MdCBF3* and negatively regulates cold tolerance. In addition, we identified a natural variation in *MdRAD5B* associated with cold tolerance in apple. These findings provide new insights into the mechanism by which MdRAD5B regulates the chromatin accessibility of *MdMYB14* through the transcription factor MdLSD1 under cold.

RESULTS

MdRAD5B is a positive regulator of cold tolerance in apple

To understand whether MdRAD5B could respond to cold, we determined the expression of *MdRAD5B* in *Malus domestica* cv. Golden Delicious treated with low temperature and found that *MdRAD5B* was slightly induced (Figure S1), indicating that *MdRAD5B* may play a role in cold stress response. To examine the biological function of MdRAD5B in cold response, we treated the previously generated *MdRAD5B* transgenic plants with cold (Song et al., 2024). Ion leakage assays showed that, irrespective of acclimation, *MdRAD5B* RNA interference (*MdRAD5B*-RNAi) plants were more sensitive to freezing stress, whereas *MdRAD5B* overexpression (*MdRAD5B*-OE) plants were more tolerant (Figure 1A–D). Freezing treatment further revealed that *MdRAD5B*-RNAi transgenic plants had a lower survival rate than GL-3, whereas *MdRAD5B*-OE plants exhibited a higher survival rate (Figure 1A, B, and E). Although cold acclimation improved overall survival rates across all genotypes, the relative order of tolerance remained unchanged (Figure 1A, B, and F). In addition, *MdRAD5B*-OE transgenic calli displayed a higher relative growth rate than WT following cold treatment (Figure S2A and B). We then assessed malondialdehyde (MDA) content and antioxidant enzyme activities. The results showed that *MdRAD5B*-OE plants accumulated less MDA than GL-3 plants, while *MdRAD5B*-RNAi plants accumulated more MDA (Figure S2C). Moreover, under cold conditions, *MdRAD5B*-OE plants showed higher peroxidase (POD) and catalase (CAT) activities than GL-3 plants, while *MdRAD5B*-RNAi plants exhibited lower activities (Figure S2D and E). Accordingly, hydrogen peroxide (H_2O_2) content in GL-3 was higher than in *MdRAD5B*-OE plants but lower than in *MdRAD5B*-RNAi plants (Figure S2F). Taken together, these results indicate that MdRAD5B is a positive regulator of cold tolerance in apple.

MdRAD5B regulates the expression of cold-responsive genes under cold

To further explore the regulatory mechanism of MdRAD5B in response to cold stress, we performed RNA sequencing (RNA-seq) using leaves from *MdRAD5B*-RNAi and GL-3 plants after cold treatment. Principal component analysis (PCA) confirmed the clustering of biological replicates (Figure S3A). RNA-seq analysis of *MdRAD5B*-RNAi and GL-3 plants revealed that a total of 1,352 differentially expressed genes (DEGs) ($|\log_2 FC|>1$, P-value<0.05) were identified in *MdRAD5B*-RNAi plants compared with GL-3 plants under cold conditions. Among the 1,352 DEGs, 951 were up-regulated while 401 were down-regulated (Figure 2A; Data S1). Gene Ontology (GO) enrichment analysis of the 1,352 DEGs showed that these DEGs were mainly involved in processes such as stimulus response, metabolic process, and phenylpropanoid biosynthetic process (Figure 2B). In addition, we generated a heatmap of the 1,927 genes responsive to cold in GL-3 plants and found that MdRAD5B preferentially inhibited cold-repressed genes and activated cold-induced genes under cold stress (Figure S3B). Furthermore, we found that 37 cold-induced genes were down-regulated in *MdRAD5B*-RNAi plants under cold conditions, whereas 83 cold-repressed genes were up-regulated (Figure 2A).

We selected 12 genes, including *MdCBF3*, *MdCCA1*, *MdLEA4-5*, *MdMYB12*, *MdMYB14*, *MdF3H*, *MdF3'H*, *MdCHI*, *MdANS*, *MdLAR*, *MdUFGT*, and *MdJAZ6*, to verify our RNA-seq data using qRT-PCR analysis. Among these genes, homologs of *MdCBF3* and *MdCCA1* are cold signaling genes that positively regulate cold tolerance (Jang et al., 2024; Zhou et al., 2011). A homolog of *MdLEA4-5* acts as a positive regulator of cold tolerance (Reyes et al., 2008). Homologs of *MdF3H*, *MdF3'H*, *MdCHI*, *MdANS*, *MdLAR*, and *MdUFGT* are involved in anthocyanin accumulation (Cong et al., 2021; Li et al., 2016; Li et al., 2022b; Li and Ahammed, 2023; Lu, 2024; Sunil and Shetty, 2022), and a homolog of *MdMYB12* can enhance cold tolerance by increasing

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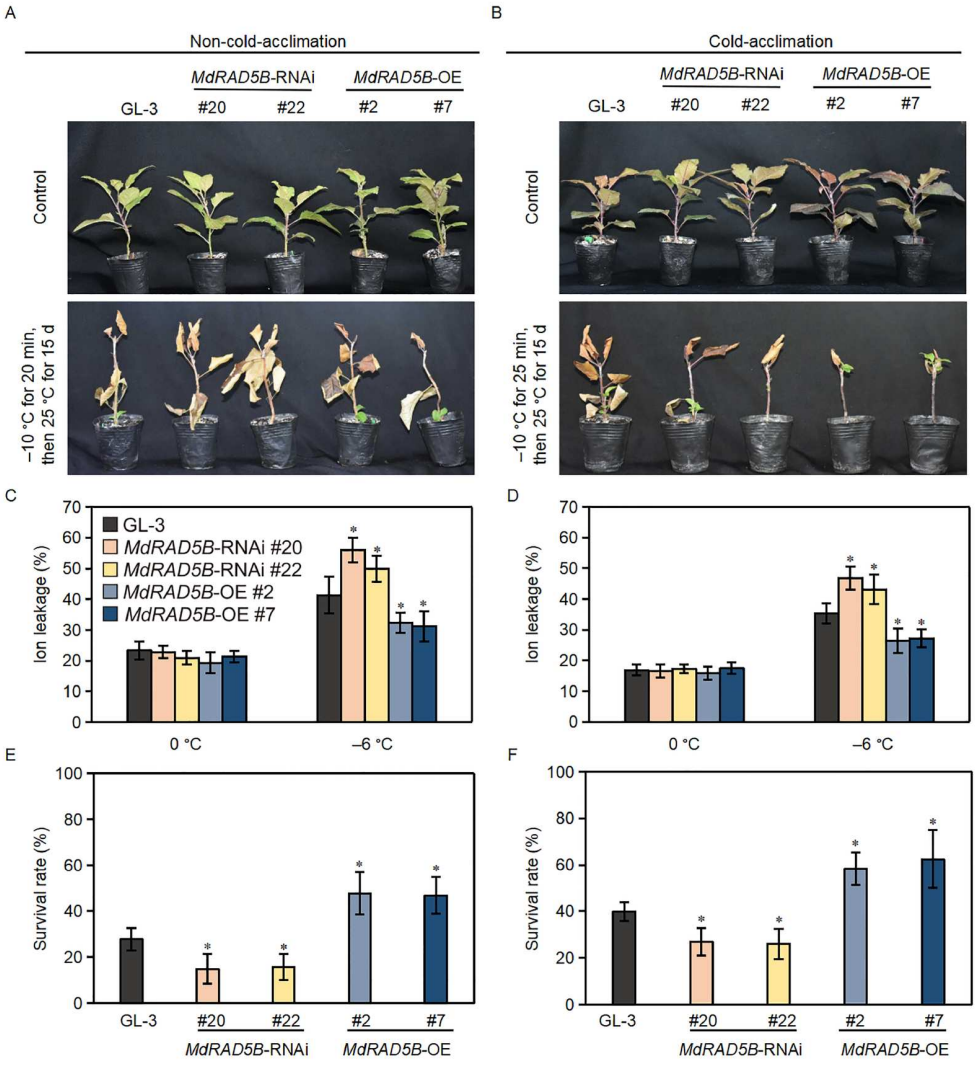


Figure 1 MdRAD5B is a positive regulator of freezing tolerance in apple. A and B, Freezing tolerance of *MdRAD5B* transgenic plants. One-month-old plants were transferred to soil for an additional month. Non-cold-acclimated GL-3 and *MdRAD5B* transgenic plants were exposed to −10°C for 20 min, while cold-acclimated plants were exposed to −10°C for 25 min. After treatment, all plants were transferred to a growth chamber for recovery. Bars=3 cm. C and D, Ion leakage of non-cold-acclimated and cold-acclimated GL-3 and *MdRAD5B* transgenic plants. One-month-old plants were transferred to soil for an additional month, and leaves were collected for an ion leakage assay. E and F, Plant survival rate of non-cold-acclimated and cold-acclimated GL-3 and *MdRAD5B* transgenic plants in A and B. Error bars indicate standard deviation (n=6 in C and D, and 36–40 in E and F, with each 12–15 plants as a biological replicate). Student's t-test was performed and statistically significant differences were indicated by *, P<0.05.

anthocyanin content under cold stress (Wang et al., 2016). However, homologs of *MdMYB14* and *MdJAZ6* are negative regulators of cold tolerance (Chen et al., 2013; Hu et al., 2017). The qRT-PCR results showed that the expression of *MdCBF3*, *MdCCA1*, *MdLEA4-5*, *MdMYB12*, *MdF3H*, *MdF3'H*, *MdCHI*, *MdANS*, *MdLAR*, and *MdUFGT* was induced in *MdRAD5B*-OE transgenic plants and was reduced in *MdRAD5B*-RNAi transgenic plants compared with GL-3 plants under cold conditions (Figure 2C). Conversely, the expression of *MdMYB14* and *MdJAZ6* was reduced in *MdRAD5B*-OE transgenic plants and induced in *MdRAD5B*-RNAi transgenic plants compared with GL-3 plants under cold conditions (Figure 2C). These results suggest that MdRAD5B enhances cold tolerance by regulating the expression of cold-responsive genes in apple.

Moreover, we measured the anthocyanin content in *MdRAD5B* transgenic plants after cold acclimation. The results showed that, consistent with the gene expression pattern, MdRAD5B promoted anthocyanin accumulation after cold treatment (Figure S4).

MdRAD5B regulates chromatin remodeling of cold-responsive genes under cold stress

MdRAD5B has been identified to regulate gene expression through its chromatin remodeling function in response to drought stress (Song et al., 2024), therefore, we performed micrococcal nuclease quantitative PCR (MNase-qPCR) to detect the chromatin accessibility of DEGs in *MdRAD5B* transgenic plants and GL-3 plants under cold stress. The results showed that under control conditions, the chromatin accessibility of the *MdANS*, *MdMYB12*, and *MdMYB14* loci showed no significant difference between GL-3 and *MdRAD5B* transgenic plants, and MdRAD5B positively regulated the chromatin accessibility of the *MdCBF3* locus. Under cold conditions, the chromatin accessibility of the *MdCBF3*, *MdMYB12*, and *MdANS* loci in GL-3 plants was higher than that in *MdRAD5B*-RNAi plants and lower than

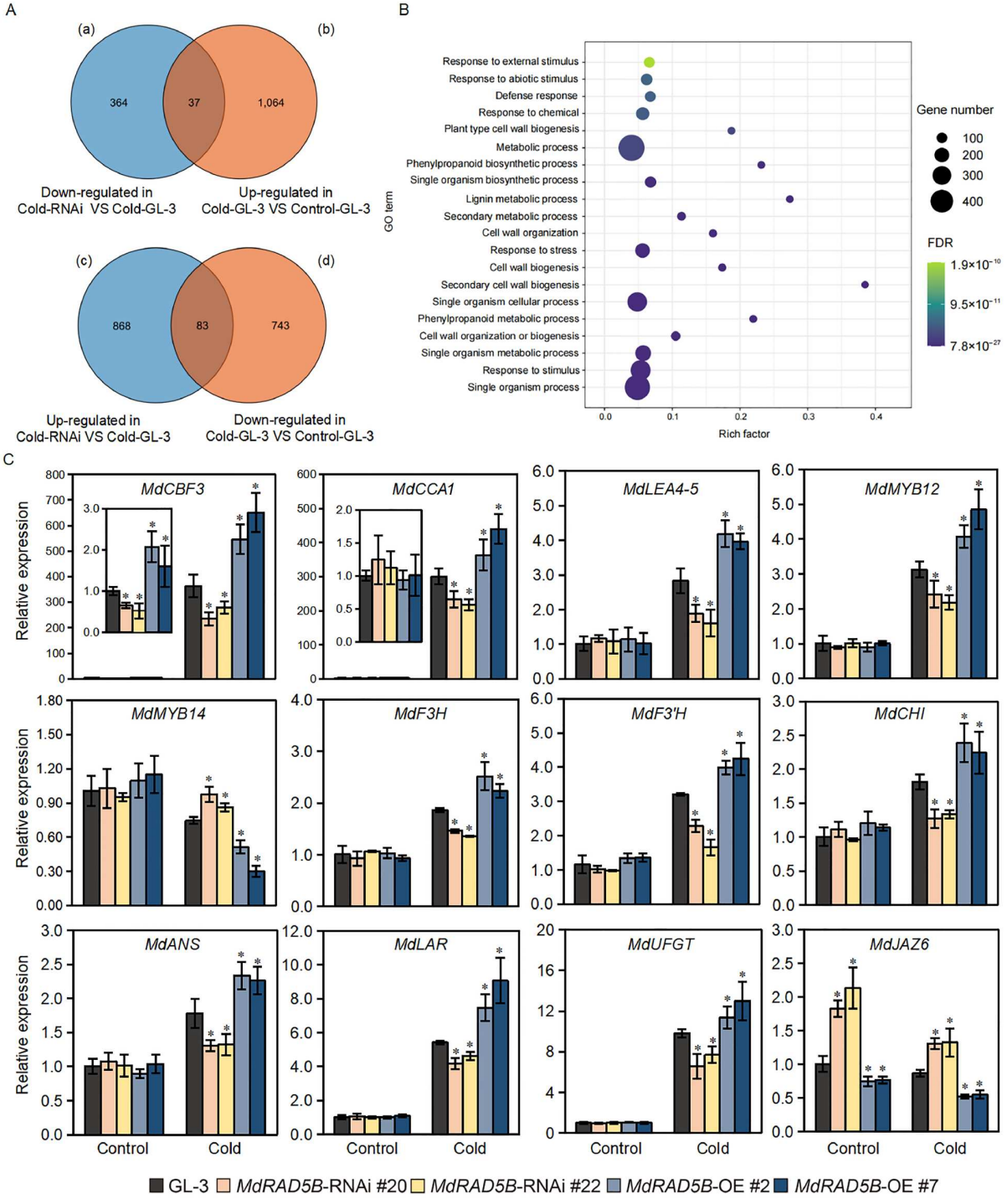


Figure 2 Regulation of cold-responsive genes by MdRAD5B in response to cold stress. A, Venn diagrams of genes under cold conditions. (a) Down-regulated genes in *MdRAD5B*-RNAi plants relative to GL-3 plants. (b) Up-regulated genes in GL-3 plants. (c) Up-regulated genes in *MdRAD5B*-RNAi plants relative to GL-3 plants. (d) Down-regulated genes in GL-3 plants. A significance threshold of $|\log_2 FC|>1$ was applied for all DEGs shown in the Venn diagrams. RNAi, *MdRAD5B*-RNAi #20. B, GO enrichment analysis of the DEGs in *MdRAD5B*-RNAi plants under cold conditions. C, Validation of RNA-seq data shown in A using qRT-PCR. Error bars indicate standard deviation ($n=3$ in C). Student's *t*-test was performed and statistically significant differences were indicated by *, $P<0.05$.

that in *MdRAD5B*-OE plants after cold stress, consistent with their expression under cold stress (Figures 2C and 3A–C). In contrast, MdRAD5B reduced the chromatin accessibility of the *MdMYB14* locus and inhibited its expression under cold stress (Figures 2C and 3D). These results indicate that MdRAD5B regulates the expression of cold-responsive genes under cold stress through its chromatin remodeling function.

MdRAD5B interacts with MdLSD1 and acts upstream of MdLSD1

To understand how MdRAD5B regulates chromatin accessibility

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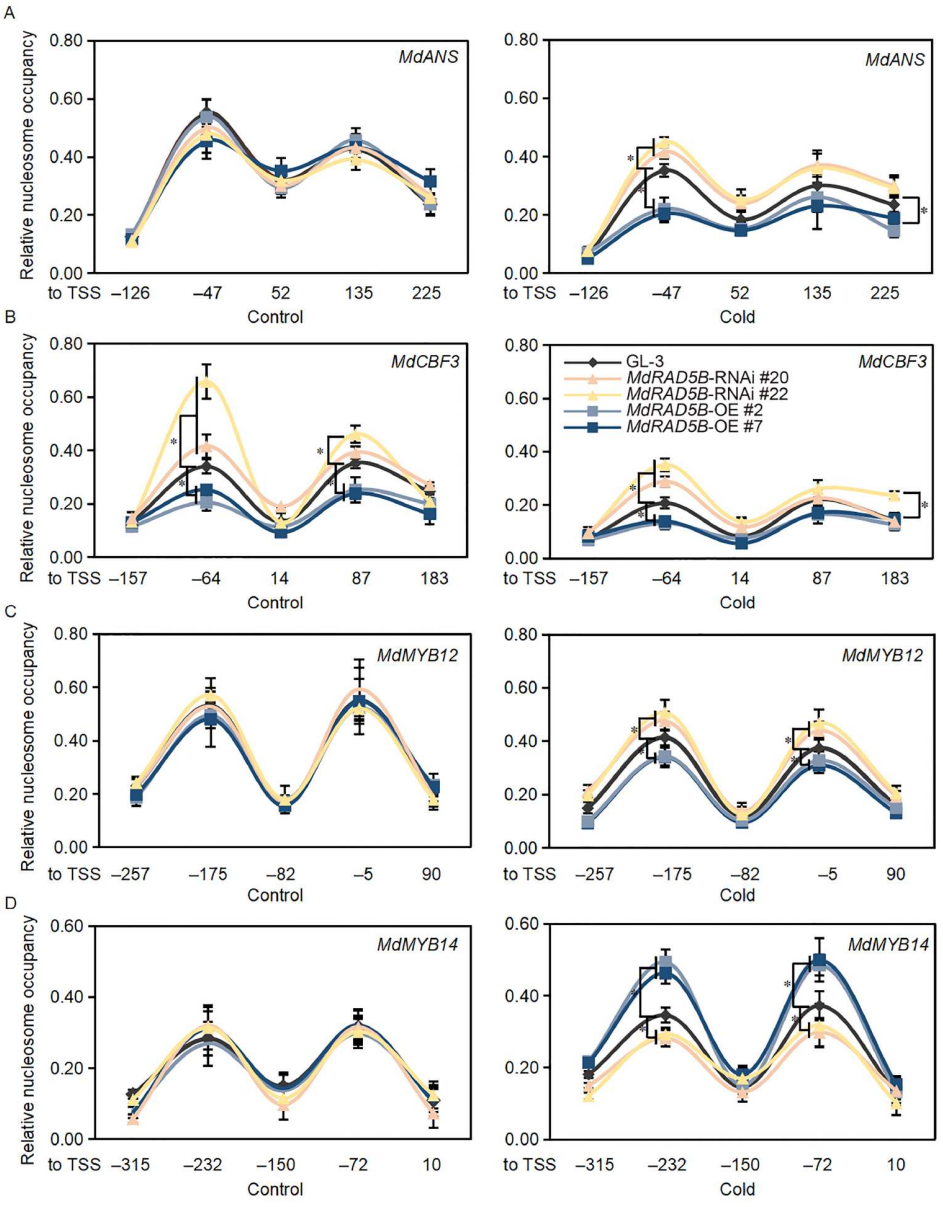


Figure 3 MdRAD5B regulates the chromatin accessibility of cold-responsive genes under cold stress. A–D, Relative nucleosome occupancy of cold-responsive genes in GL-3 and *MdRAD5B* transgenic plants. The number on the *x*-axis denotes distance (bp) from the TSS (0 bp). Error bars indicate standard deviation (n=3). Student's *t*-test was performed and statistically significant differences were indicated by *, P<0.05.

of genes, we performed a yeast two-hybrid (Y2H) screening and identified one potential interacting protein, LESION SIMULATING DISEASE 1 (MdLSD1). Y2H analysis showed that MdRAD5B interacted with MdLSD1 in yeast cells (Figure 4A). Split-luciferase complementation (Split-LUC) and co-immunoprecipitation (Co-IP) assays further illustrated the *in vivo* interaction (Figure 4B and C). When cLUC-MdRAD5B and MdLSD1-nLUC were co-expressed in tobacco leaves, the firefly luciferase activity could be detected, whereas no signal was detected when either plasmid was co-expressed with the empty vector (Figure 4B). In addition, using an α-Flag antibody to immunoprecipitate MdRAD5B-Flag, Myc-MdLSD1 was detected with an α-Myc antibody in tobacco leaves when MdRAD5B-Flag and Myc-MdLSD1 were co-expressed (Figure 4C). Moreover, we performed Y2H assays using truncated MdRAD5B and MdLSD1. The results showed that the HIRAN domain, but not the RING domain, of MdRAD5B interacted with MdLSD1 (Figure 4D and E). Consistently, MdRAD5B did not degrade MdLSD1 under cold stress (Figure S5).

Further qRT-PCR analysis showed that *MdLSD1* was induced under cold stress (Figure S6A). To investigate the biological function of MdLSD1 in cold response, we generated transgenic apple calli by overexpressing an antisense fragment of the *MdLSD1* coding sequence (CDS) (*MdLSD1*-anti) (Figure S6B and C). After cold treatment, *MdLSD1*-anti calli exhibited a lower relative growth rate than WT (Figure S6D and E), indicating that MdLSD1 plays a positive role in response to cold stress in apple.

To investigate the genetic relationship between *MdRAD5B* and *MdLSD1*, we generated double transgenic calli through overexpression of *MdRAD5B* in *MdLSD1*-anti calli (*MdRAD5B*-OE/

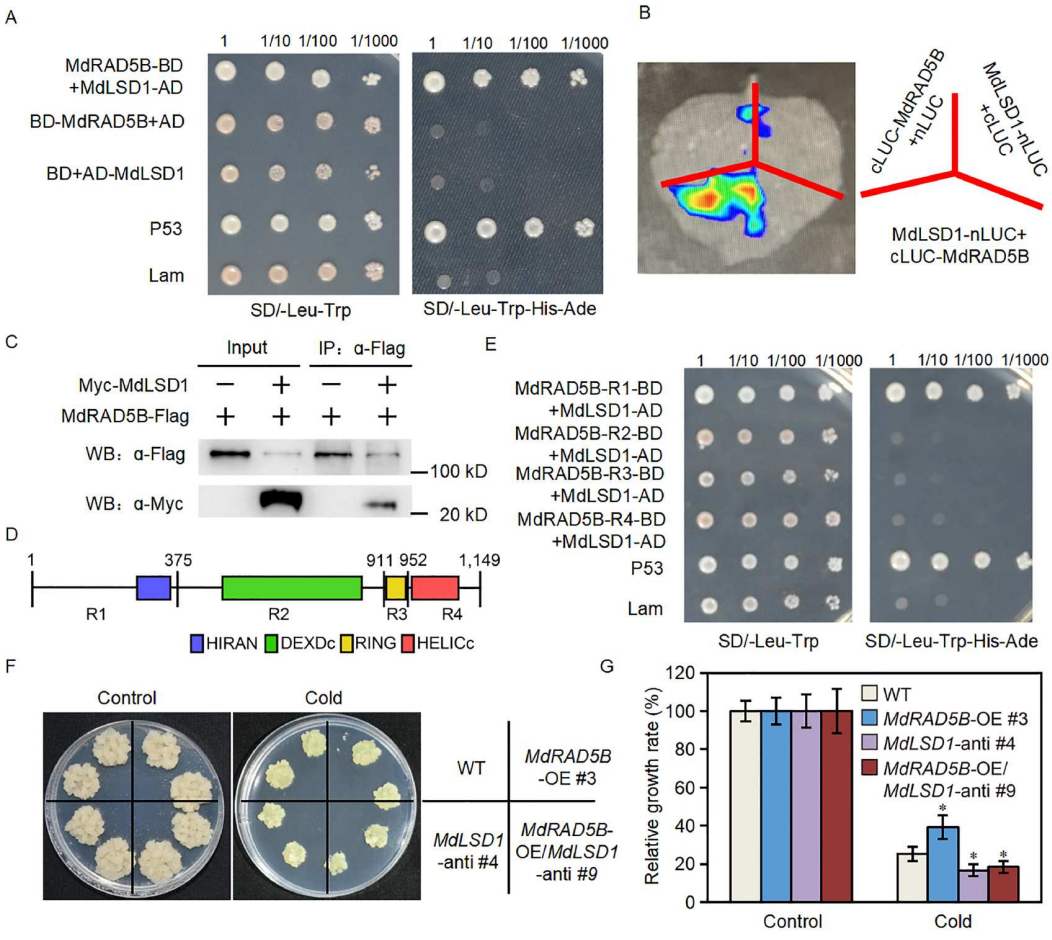


Figure 4 MdRAD5B interacts with MdLSD1 and acts upstream of MdLSD1. A, Y2H assays confirmed the interaction between MdRAD5B and MdLSD1, using fusions of the full-length proteins with the GAL4 binding domain (MdRAD5B) or activation domain (MdLSD1). B, Split-luciferase assays showing MdRAD5B-MdLSD1 interaction in tobacco. MdLSD1-nLUC and MdRAD5B-cLUC fusions were used, and LUC signals were assayed 3 d post-infiltration. C, Interaction of MdRAD5B and MdLSD1 determined by Co-IP assay. D, Domain structure of MdRAD5B. Based on the four domains of MdRAD5B, the protein was divided into four regions and subjected to Y2H assays. R1, 1-375AA. R2, 376-850AA. R3, 851-970AA. R4, 971-1149AA. R, region. E, Interaction between truncated MdRAD5B and MdLSD1 determined by Y2H assays. F, Morphological characteristics of WT, *MdLSD1*-anti #4, *MdRAD5B*-OE #3, and *MdRAD5B*-OE/*MdLSD1*-anti #9 transgenic apple calli under control or cold conditions. Under control conditions, transgenic calli were cultured at 22°C for 2 weeks. Under cold conditions, transgenic calli were cultured at 22°C for 1 week and then transferred to 4°C for an additional week. G, The relative growth rate of the apple calli shown in F. Error bars indicate standard deviation (n=5 in G). Student's t-test was performed and statistically significant differences were indicated by *, P<0.05.

MdLSD1-anti) (Figure S7A–D). The results showed that overexpression of *MdRAD5B* partially restored the reduced relative growth rate of *MdLSD1*-anti calli (Figure 4F and G; Figure S7E and F), indicating that *MdRAD5B* acts upstream of *MdLSD1* in response to cold stress.

MdLSD1 is a transcription factor that recognizes the TGATAA motif

Multi-sequence alignment analysis reveals that the MdLSD1 protein and AtLSD1 protein belong to the GATA1-like family of zinc-finger transcription factors and share a high degree of similarity in three conserved LSD1 zinc finger domains (Figure S8A). The three zinc finger domains in MdLSD1 exhibited a $CX_2CX_{14}CX_2C$ zinc finger structure, similar to the B-box domain of AtCO, which has the ability to bind to DNA (Figure S8B) (Omichinski et al., 1993; Takatsuji, 1998), indicating that MdLSD1 may be able to bind to DNA. The Y2H analysis showed that MdLSD1 does not have self-activation activity (Figure S8C), consistent with the fact that AtLSD1 is a transcriptional repressor (Czarnocka et al., 2017). We then hypothesized that MdLSD1 is involved in the inhibition of cold-responsive genes by MdRAD5B under cold stress. Among the verified and repressed DEGs of MdRAD5B, *MdMYB14* was found to be inhibited by MdLSD1 under cold conditions in a dual-luciferase reporter (Dual-LUC) assay, while *MdJAZ6* was not (Figure 5A; Figure S9A). To further verify the direct binding of MdLSD1 to the *MdMYB14* promoter, we performed a chromatin immunoprecipitation quantitative PCR (ChIP-qPCR) assay, an electrophoretic mobility shift assay (EMSA) assay, and a DNA affinity purification quantitative PCR (DAP-qPCR) assay. ChIP-qPCR assay using *MdLSD1*-OE transgenic calli (Figure S10) demonstrated that MdLSD1 could bind to the P1 and P2 regions of the *MdMYB14* promoter (Figure 5B and C). EMSA (Figure S11A) and DAP-qPCR analysis (Figure 5D) showed that MdLSD1 only bound to the P1 region of the *MdMYB14* promoter.

To explore the specific motif recognized by MdLSD1, we truncated the P1 region (247 bp) into five fragments including

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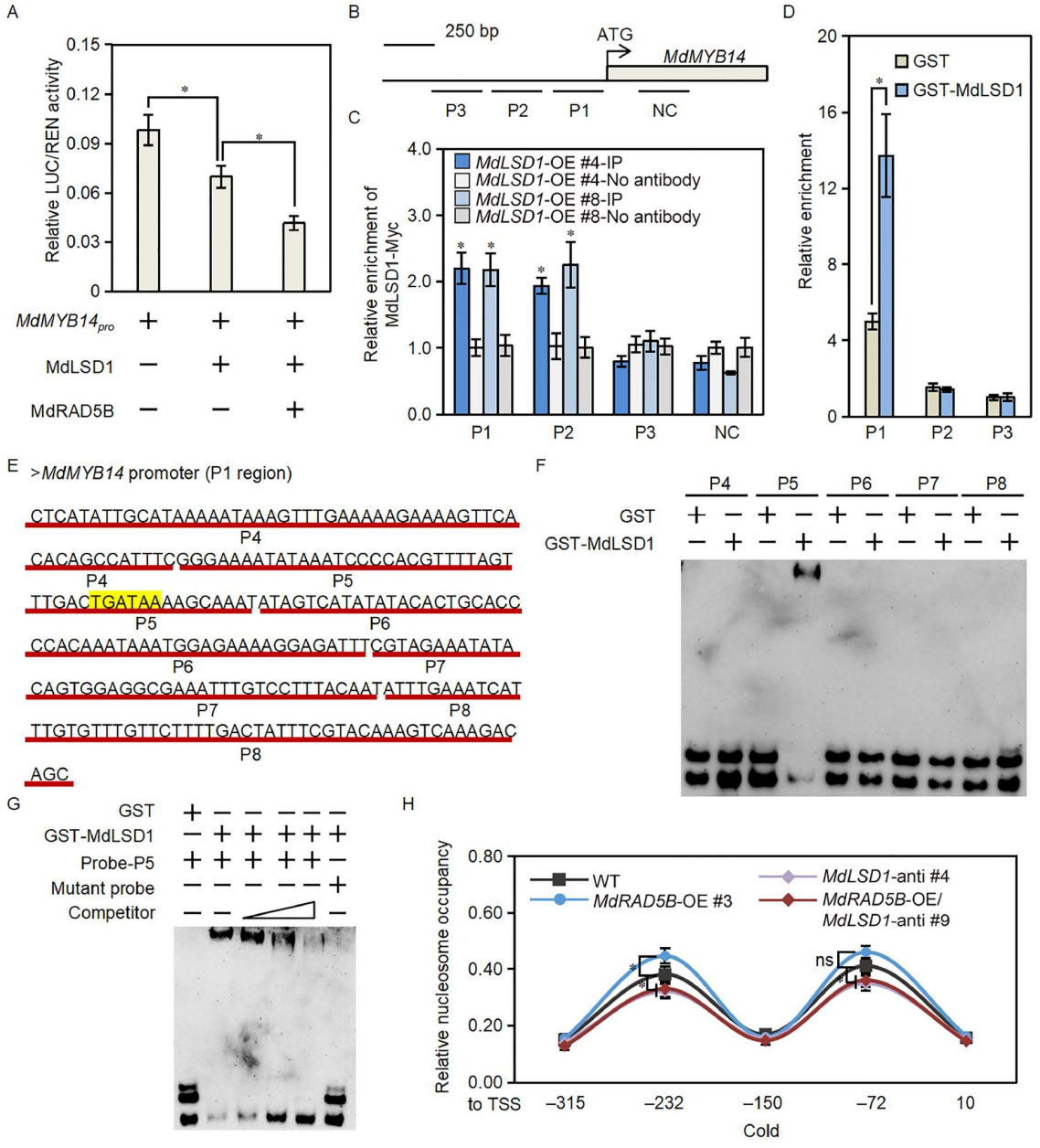


Figure 5 MdRAD5B regulates the chromatin accessibility of the *MdMYB14* locus through the transcription factor MdLSD1. A, Dual-LUC assay showing transcriptional regulation of the *MdMYB14* promoter by MdLSD1 and MdRAD5B under cold conditions. B, Diagram of *MdMYB14* promoter regions. The *MdMYB14* promoter was divided into 3 regions, including P1, P2, and P3. NC, negative control. C, ChIP-qPCR analysis of MdLSD1 binding to the P1 and P2 regions of the *MdMYB14* promoter under cold conditions. D, DAP-qPCR analysis of MdLSD1 binding to the P1 region of the *MdMYB14* promoter. E, Sequence of the P1 region (247 bp) of the *MdMYB14* promoter. The P1 region was truncated into five fragments, including P4 (53 bp), P5 (47 bp), P6 (50 bp), P7 (50 bp), and P8 (47 bp). F, EMSA of MdLSD1 binding to the P5 fragment of the *MdMYB14* promoter. G, EMSA analysis of MdLSD1 binding to the TGATAA motif of the *MdMYB14* promoter. The sequence of the P5 fragment containing the TGATAA motif was used as the probe, and the sequence of the P5 fragment with the TGATAA motif mutated to GTCGCC was used as the mutant probe. H, Relative nucleosome occupancy of *MdMYB14* in WT, *MdLSD1*-anti #4, *MdRAD5B*-OE #3, and *MdRAD5B*-OE/*MdLSD1*-anti #9 transgenic apple calli under cold conditions. The number on the *x*-axis denotes distance (bp) from the TSS (0 bp). Error bars indicate standard deviation (n=8 in A, n=3 in C, D, and H). Student's t-test was performed and statistically significant differences were indicated by *, P<0.05.

P4 (53 bp), P5 (47 bp), P6 (50 bp), P7 (50 bp), and P8 (47 bp) (Figure 5E). The EMSA analysis showed that MdLSD1 could bind to the P5 fragment of the *MdMYB14* promoter (Figure 5F). Previous research showed that GATA1 could bind to the TGATAA motif (Omichinski et al., 1993; Zhang et al., 2016a). Since LSD1 belongs to the GATA1-like transcription factor family (Figure S8A), and the P5 fragment contains a TGATAA motif (Figure 5E), we hypothesized that MdLSD1 might bind to the TGATAA motif in the P5 fragment. We then mutated the TGATAA motif in the P5 fragment to GTCGCC and found that MdLSD1 no longer bound to the mutant P5 fragment (Figure 5G). In addition, subcellular localization analysis showed that the MdLSD1 protein was localized in the nucleus and cytoplasm (Figure S8D). Therefore, MdLSD1 is an apple transcription factor whose binding motif is TGATAA. Additionally, consistent with the Dual-LUC assay, MdLSD1 did not bind to the *MdJAZ6*

promoter or regulate its expression under cold conditions (Figure S9B–D).

MdRAD5B reduces the chromatin accessibility and inhibits the expression of *MdMYB14* through MdLSD1

To verify whether the reduction of *MdMYB14* chromatin accessibility and the inhibition of its expression by MdRAD5B under cold stress are associated with MdLSD1, we conducted MNase-qPCR in WT, *MdRAD5B*-OE, *MdLSD1*-anti, and *MdRAD5B*-OE/*MdLSD1*-anti calli under control and cold conditions. Under control conditions, the chromatin accessibility of the *MdMYB14* locus showed no significant difference in WT and transgenic calli. Under cold conditions, while overexpression of *MdRAD5B* reduced the chromatin accessibility of the *MdMYB14* locus, the suppression of *MdLSD1* in *MdRAD5B*-OE calli interfered with the regulation of the *MdMYB14* locus by MdRAD5B (Figure 5H; Figure S11B). Consistent with these results, qRT-PCR analysis showed that *MdRAD5B*-OE calli exhibited inhibited *MdMYB14* expression, whereas *MdLSD1*-anti and *MdRAD5B*-OE/*MdLSD1*-anti calli displayed comparable levels to the WT under cold conditions (Figure S11C). Furthermore, MdRAD5B enhanced the inhibition of *MdMYB14* expression by MdLSD1 (Figure 5A). Taken together, these results suggest that the reduction of *MdMYB14* chromatin accessibility and the inhibition of its expression by MdRAD5B under cold stress are mediated by the transcription factor MdLSD1.

MdMYB14 negatively regulates cold tolerance in apple

We then explored the biological function of MdMYB14 in apple response to cold stress. The expression of *MdMYB14* was inhibited under cold stress (Figure 6A), indicating that *MdMYB14* is involved in apple cold response. Multi-sequence alignment analysis revealed that MdMYB14 shared a high degree of similarity in its two consecutive conserved SANT domains with AtMYB14; therefore, it is predicted to act as an R2R3 MYB transcription factor (Figure 6B; Figure S12A). Subcellular localization showed that the MdMYB14 protein was localized in the nucleus (Figure S12B). Subsequently, we generated *MdMYB14* overexpression (*MdMYB14*-OE) transgenic calli to investigate the function of MdMYB14 in regulating cold tolerance in apple (Figure S13). Under cold conditions, the relative growth rate of *MdMYB14*-OE transgenic calli was lower than that of the WT (Figure 6C and D), indicating that MdMYB14 plays a negative role in apple cold tolerance.

MdCBF3 is a direct target of MdMYB14

Although the molecular mechanism by which MYB14 regulates target gene expression remains unclear, studies have revealed that AtMYB15, an R2R3 MYB transcription factor with high similarity to the AtMYB14 protein in the two conserved SANT domains, can bind to the "GTGGT" site in the promoter of *AtCBF3* (Wang et al., 2023). Therefore, we hypothesized that MdMYB14 might have a similar regulatory mechanism. In apple, *MdCBF3* has two GTGGT motifs in its promoter (Figure 7A). To verify whether *MdCBF3* is a target gene of MdMYB14, we first performed a Dual-LUC assay. The results showed that co-injection of MdMYB14 and the *MdCBF3* promoter reporter significantly reduced the expression of the LUC reporter gene (Figure 7B). Further qRT-PCR results showed that under cold stress, overexpression of *MdMYB14* significantly inhibited the expression of *MdCBF3* (Figure 7C). We then conducted a ChIP-qPCR, and the results showed that MdMYB14 could bind to the P1 region of the *MdCBF3* promoter, which contains two GTGGT motifs (Figure 7A and D; Figure S13). Subsequently, we performed an EMSA to further verify the binding of MdMYB14 to the two sites within the P1 region. Results showed that MdMYB14 could bind to both fragments that contain the GTGGT motif (Figure 7E and F). Collectively, these results identified *MdCBF3* as a direct target of MdMYB14.

MdMYB14 plays a critical role in cold response

To provide a broader view of the regulatory scope of MdMYB14, we performed RNA-seq using WT and *MdMYB14*-OE transgenic calli under cold conditions. PCA confirmed the clustering of biological replicates (Figure S14A). Comparative analysis revealed 5,590 cold-responsive genes in WT calli and 5,358 DEGs ($|\log_2 FC|>1$, P-value<0.05) in *MdMYB14*-OE calli relative to WT calli under cold conditions (Figure 7G; Figure S14B and Data S2). We also generated a heatmap to visualize core cold-responsive DEGs across samples (Figure S14C). Among these 5,358 DEGs, 3,477 were down-regulated while 1,881 were up-regulated, suggesting that MdMYB14 primarily functions as a transcriptional repressor under cold conditions. Moreover, to identify potential targets of MdMYB14, we analyzed the enrichment of the GTGGT motif within the 1,000 bp upstream regions of the transcription start sites of DEGs in *MdMYB14*-OE calli compared with WT calli under cold conditions, leading to the identification of 4,835 putative target genes. Among these, 1,785 genes are known to be involved in the cold stress response in apple (Figure 7H), underscoring the critical role of MdMYB14 in the cold response network.

MdRAD5B^{T594A} exhibits weaker activity than MdRAD5B in responses to cold stress in apple

Given that MdRAD5B positively regulates cold tolerance in apple, we are interested in whether *RAD5B* is associated with cold tolerance in an ecological context. We determined ion leakage in annual dormant branches from 109 *Malus* accessions (Li et al., 2022c) and selected 14 (8 cold-tolerant and 6 cold-sensitive) for *RAD5B* gene characterization (Table S1). The qRT-PCR analysis revealed that while the expression of the *RAD5B* gene was significantly increased under cold stress, there was no difference in *RAD5B* expression among different accessions (Figure S15A). However, sequence analysis of the CDS of *RAD5B* from various accessions identified a variation from A to G at the 1,780 position of the *RAD5B* CDS (*RAD5B*^{CDS1780}), resulting in an amino acid substitution from threonine (T) to alanine (A) at the 594 position of the RAD5B protein (Figure S15B).

The 594T position of the MdRAD5B protein lies adjacent to the 562K position, a key ATPase activity site of MdRAD5B (Knizewski et al., 2008; Song et al., 2024; Sung et al., 1988), indicating that this variation might affect the ATPase activity of RAD5B. Therefore, we detected the ATPase activity of MdRAD5B^{T594A} and found that MdRAD5B^{T594A} exhibited weaker ATPase activity than MdRAD5B (MdRAD5B^{594T}) (Figure 8A). We also generated transgenic calli expressing *35S:*

https://doi.org/10.1007/s11427-025-3445-5

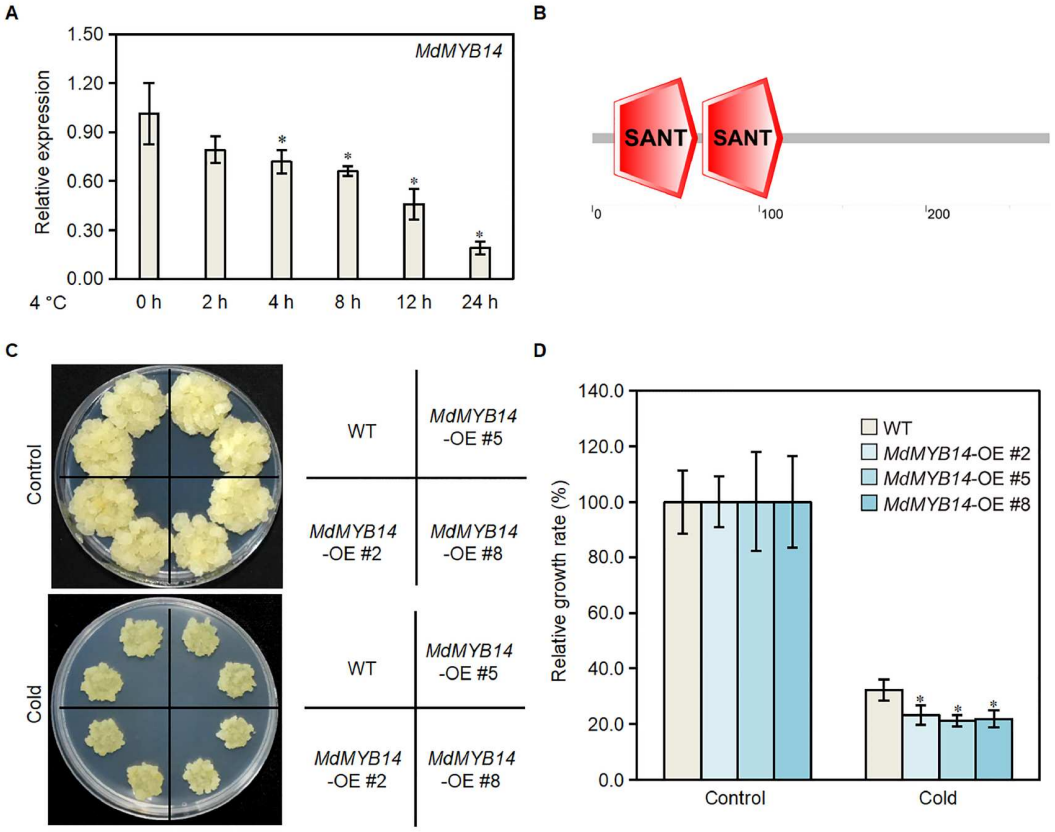


Figure 6 MdMYB14 negatively regulates cold tolerance in apple. A, Time-course expression of *MdMYB14* in *Malus domestica* in response to cold stress. B, Structure analysis of the MdMYB14 protein. C, Morphological characteristics of WT and *MdMYB14*-OE #2, #5, and #8 transgenic apple calli under control or cold conditions. Under control conditions, transgenic calli were cultured at 22°C for 2 weeks. Under cold conditions, transgenic calli were cultured at 22°C for 1 week and then transferred to 4°C for an additional week. D, The relative growth rate of the apple calli shown in C. Error bars indicate standard deviation (n=3 in A, n=5 in C and D). Student's t-test was performed and statistically significant differences were indicated by *, P<0.05.

MdRAD5B^{T594A} (*MdRAD5B*^{T594A}-OE) (Figure S16). Under cold conditions, although *MdRAD5B*-OE (*MdRAD5B*^{594T}-OE) and *MdRAD5B*^{T594A}-OE calli were more tolerant than WT, the relative growth rate of *MdRAD5B*-OE calli remained higher than that of *MdRAD5B*^{T594A}-OE calli (Figure 8B and C).

Furthermore, we examined the chromatin accessibility of the *MdCBF3*, *MdANS*, *MdMYB12*, and *MdMYB14* loci in WT, *MdRAD5B*^{T594A}-OE, and *MdRAD5B*-OE calli using MNase-qPCR. Under control conditions, the chromatin accessibility of the *MdANS*, *MdMYB12*, and *MdMYB14* loci, but not the *MdCBF3* locus, showed no significant difference in WT and transgenic calli. Under cold conditions, although overexpression of both *MdRAD5B* and *MdRAD5B*^{T594A} increased chromatin accessibility at the *MdANS*, *MdCBF3*, and *MdMYB12* loci, *MdRAD5B* overexpression resulted in a more pronounced increase than *MdRAD5B*^{T594A} did. In contrast, overexpression of both *MdRAD5B* and *MdRAD5B*^{T594A} decreased chromatin accessibility of the *MdMYB14* locus under cold conditions, while overexpression of *MdRAD5B* reduced more chromatin accessibility of the *MdMYB14* locus than overexpression of *MdRAD5B*^{T594A} did (Figure S17). Meanwhile, the gene expression results were consistent with the chromatin accessibility data (Figure S18). Collectively, these results demonstrate that the amino acid sequence variance of RAD5B at the 594 position affects the chromatin accessibility and expression patterns of related genes under cold stress.

The *RAD5B*^{CDS1780} SNP is associated with cold tolerance and has undergone artificial selection

To further verify whether the variation at position 1,780 of the *MdRAD5B* CDS (a G/A SNP) is associated with cold tolerance in apple, we performed derived cleaved amplified polymorphic sequences (dCAPS) analysis on the same 109 accessions used for ion-leakage measurements. The results showed that accessions with the A/A genotype exhibited significantly stronger cold tolerance than those with the A/G and G/G genotypes (Figure 8D; Figure S19 and Table S1). We then conducted a genotypic frequency analysis of the *RAD5B*^{CDS1780} locus. We divided 109 accessions into relatively cold-tolerant (n=62) and relatively cold-sensitive (n=47) groups based on the threshold of electrolyte leakage rate at 42.5% (Table S1). The analysis revealed that nearly all relatively cold-tolerant accessions carried the A/A genotype (n=58), with only four accessions exhibiting the A/G (n=3) or G/G (n=1) genotype. In contrast, the A/G (n=10) and G/G (n=8) genotypes were present at relatively higher frequencies in the cold-sensitive group (Figure 8E). These results suggest that the *RAD5B*^{CDS1780} SNP is associated with cold tolerance.

Moreover, to determine whether this SNP locus is under artificial or natural selection, we classified the 109 apple accessions into wild accessions and cultivated varieties based on previous studies (Li et al., 2022c). Statistical analysis showed that the accessions comprised 36 wild accessions and 73

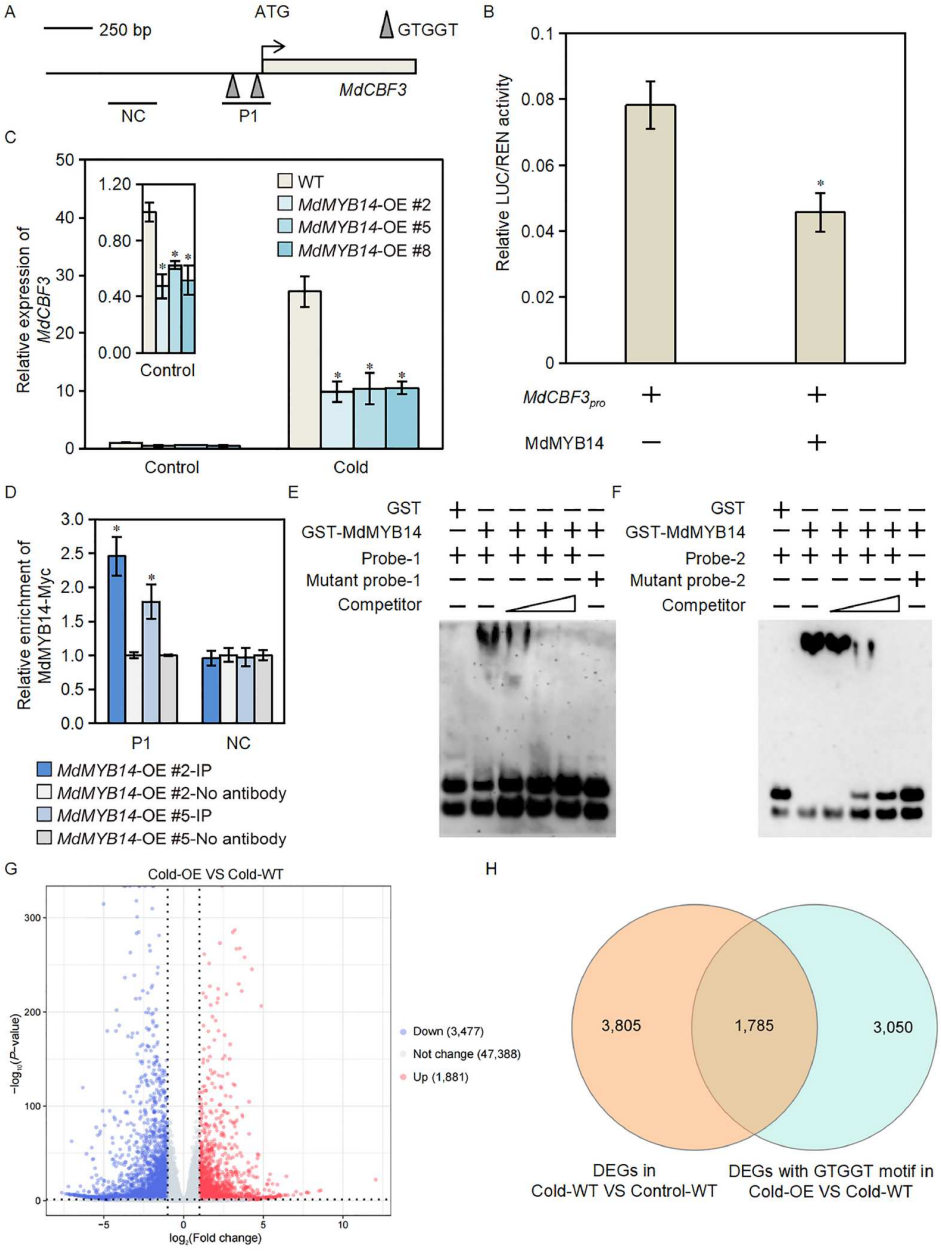


Figure 7 MdMYB14 can bind to the promoter of *MdCBF3*. A, Diagram of the *MdCBF3* promoter. Fragment P1 contains the GTGGT element. B, Dual-LUC assay showing transcriptional regulation of *MdCBF3* promoters by MdMYB14 under cold conditions. C, Relative expression level of *MdCBF3* in *MdMYB14*-OE transgenic apple calli under control or cold conditions. D, ChIP-qPCR analysis of MdMYB14 binding to the promoter of *MdCBF3* under cold conditions. E and F, EMSA analysis of MdMYB14 binding to the promoter of *MdCBF3*. G, Volcano plot showing up-regulated and down-regulated genes in *MdMYB14-OE* calli relative to WT calli under cold conditions. OE, *MdMYB14*-OE #2. A significance threshold of $|\log_2 FC|>1$ was applied for all DEGs shown in the volcano plot. H, Venn diagram showing the number of DEGs that respond to cold stress and are regulated by MdMYB14 via the GTGGT motif under cold stress. The 1,000 bp region upstream of the transcription start sites of DEGs was used as the target sequence for GTGGT motif enrichment analysis. Error bars indicate standard deviation (n=8 in B, n=3 in C and D). Student's t-test was performed and statistically significant differences were indicated by *, P<0.05.

cultivated varieties (Table S1). Further genotypic frequency analysis revealed that the A/A genotype was widely present in both wild species and cultivated varieties, whereas the A/G and G/G genotypes were mostly confined to wild accessions (Figure 8F). These findings indicate that this SNP locus has undergone artificial selection.

DISCUSSION

In the context of global warming and the increasing frequency of extreme weather events, cold stress has become a major environmental constraint that affects plant growth and productivity (Ritonga and Chen, 2020). Apple (*Malus domestica*), as a globally important fruit crop, is highly susceptible to cold stress, which adversely affects its growth, yield, and fruit quality (Feng et al., 2012). Although progress has been made in understanding apple cold tolerance (An et al., 2021; Hou et al., 2025; Shen et al., 2023; Xie et al., 2018), the molecular mechanisms remain incompletely elucidated due to the high genomic heterozygosity and genetic complexity of this species. In this study, we identified

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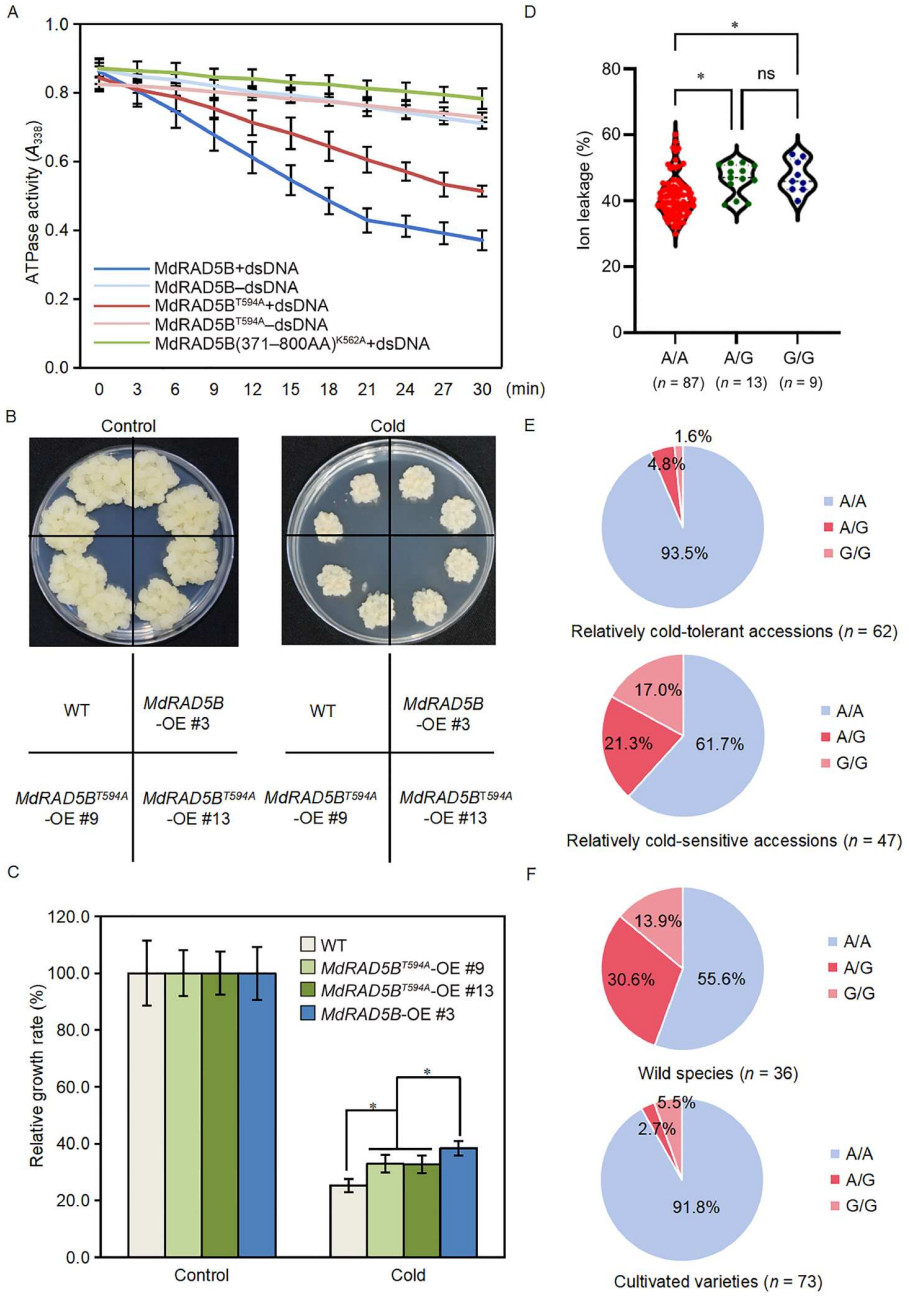


Figure 8 The *RAD5B*^{CDS1780} SNP is associated with cold tolerance and has undergone artificial selection. A, ATPase activity of MdRAD5B (MdRAD5B^{594T}) and MdRAD5B^{T594A}. ATPase activity was measured spectrophotometrically with or without dsDNA. MdRAD5B(371-800AA)^{K562A}, a mutant of MdRAD5B with its Lys (K) at the 562 position mutated to Ala (A). B, Morphological characteristics of WT, *MdRAD5B*-OE, and *MdRAD5B*^{T594A}-OE transgenic apple calli under control or cold conditions. Under control conditions, transgenic calli were cultured at 22°C for 2 weeks. Under cold conditions, transgenic calli were cultured at 22°C for 1 week and then transferred to 4°C for an additional week. C, The relative growth rate of the WT, *MdRAD5B*-OE, and *MdRAD5B*^{T594A}-OE transgenic apple calli under control or cold conditions. D, Haplotype analysis of *RAD5B*^{CDS1780} based on the ion leakage and the dCAPS results of 109 *Malus* accessions (n=87 for haplotype A/A, n=13 for haplotype A/G, n=9 for haplotype G/G). E, Genotype distributions of *RAD5B*^{CDS1780} in relatively cold-tolerant accessions and relatively cold-sensitive accessions. F, Genotype distributions of *RAD5B*^{CDS1780} in wild accessions and cultivated varieties. Error bars indicate standard deviation (n=3 in B, n=5 in D). Student's t-test was performed and statistically significant differences were indicated by *, P<0.05.

MdRAD5B as a positive regulator of cold tolerance in apple. We demonstrated that MdRAD5B modulated the expression of cold-responsive genes through chromatin remodeling. Additionally, we discovered that MdRAD5B interacted with MdLSD1, a positive regulator of cold tolerance, and that this interaction contributed to the reduction of the chromatin accessibility of the *MdMYB14* locus. Furthermore, we demonstrated that MdMYB14 negatively regulated cold tolerance by directly binding to the promoter region of *MdCBF3*, thereby reducing its expression. Moreover, a natural variation associated with apple cold tolerance was identified. These findings collectively outline a previously uncharacterized regulatory module in which MdRAD5B, in concert with MdLSD1, fine-tunes the transcriptional dynamics of cold-responsive genes through epigenetic and transcriptional mechanisms, thereby enhancing cold adaptation in apple.

Chromatin remodeling factors play a crucial role in plant

responses to abiotic stresses. When plants are exposed to abiotic stresses such as low temperature, these factors modify chromatin architecture to modulate transcriptional programs, thereby influencing physiological and biochemical processes that collectively enhance stress resilience (Song et al., 2021). For instance, the CHD-family protein PICKLE facilitates cold acclimation by maintaining an open chromatin state and cooperating with the transcription factor ICE1 to activate *CBF* expression (Yang et al., 2019). In rice, OsCHR11 promotes chromatin relaxation at gene promoters to improve cold tolerance (Li et al., 2024). Our previous work has demonstrated that MdRAD5B regulates drought-responsive genes by altering chromatin accessibility (Song et al., 2024). Given these conserved mechanisms, we hypothesized that MdRAD5B might similarly function in cold stress responses in apple. RNA-seq analysis revealed that MdRAD5B influenced the expression of genes associated with cold response (Figure 2). Further MNase-qPCR assays confirmed that MdRAD5B modulated chromatin accessibility at key cold-responsive genes (Figure 3), indicating its role in the epigenetic regulation of apple cold tolerance. Notably, not all DEGs correlated with changes in chromatin accessibility, suggesting that some regulatory effects may be indirect or involve temporal delays between chromatin remodeling and transcriptional output (Potter et al., 2018).

Understanding the targeting mechanisms of chromatin remodelers is essential, as different remodeler families perform specialized functions despite lacking intrinsic DNA-sequence recognition capability. Rather than binding DNA directly, chromatin remodelers rely on TFs for genomic localization (Clapier et al., 2017). Conversely, remodelers play a critical role in establishing accessible chromatin configurations at enhancer regions, thereby facilitating TF binding (Huang et al., 2025). Most TFs recruit chromatin remodelers through *cis*-regulatory motifs situated in nucleosome-free linker DNA. Notably, pioneer transcription factors represent a specialized class capable of binding target sites within compacted chromatin, subsequently enabling the recruitment of additional TFs and remodeling complexes (Clapier et al., 2017). In this study, we identified MdLSD1, a transcription factor, as an interaction partner and functional downstream effector of MdRAD5B during cold stress response in apple (Figure 4F and G). Through phenotypic analysis of transgenic calli, we demonstrated that MdRAD5B reduced the chromatin accessibility of the *MdMYB14* locus and thereby decreased the expression of *MdMYB14 via* MdLSD1 (Figures 3 and 5; Figure S11). These findings suggest that MdLSD1 contributes to the target specificity of MdRAD5B, enabling precise regulation of downstream genomic regions. This not only elucidates a key aspect of MdRAD5B-mediated cold tolerance but also expands our understanding of the targeting mechanisms employed by chromatin remodelers in plants.

LSD1 functions as a key negative regulator of cell death, integrating multiple signaling pathways in response to diverse biotic and abiotic stresses (Karpiński et al., 2013). Nevertheless, the precise molecular mechanisms and cellular pathways involving LSD1 remain poorly characterized. In this study, structural analysis reveals that MdLSD1 contains three zinc-finger domains, suggesting a potential role in DNA binding (Figure S8A and B) (Omichinski et al., 1993; Takatsuji, 1998). Consistent with this, Dual-LUC, ChIP-qPCR, EMSA, and DAP-qPCR assays demonstrate that MdLSD1 acts as a transcription factor and directly binds to the TGATAA motif of the *MdMYB14* promoter, it recruits MdRAD5B to reduce the chromatin accessibility and inhibit the transcription of *MdMYB14* (Figure 5; Figure S11). Beyond its role as a transcription factor, LSD1 has also been reported to function as a scaffold protein. For instance, LSD1 interacts with the bZIP10 TF, and this interaction is essential for the regulation of runaway cell death in both R-gene-mediated and basal immune responses (Kaminaka et al., 2006). Additionally, LSD1 can be recruited by the transcription factor GOLDEN2-LIKE 1 (GLK1) and subsequently inhibit GLK1 DNA-binding activity (Li et al., 2022a). In our experiments, while MdLSD1 was enriched at both the P1 and P2 regions of the *MdMYB14* promoter *in vivo* (Figure 5B and C), it exhibited direct binding only to the P1 region *in vitro* (Figure 5D; Figure S11A). This discrepancy suggests that MdLSD1 may be recruited to the P2 region through protein-protein interactions with other transcription factors, highlighting a potential scaffolding role in regulating *MdMYB14* expression.

MYB transcription factors play important roles in responding to abiotic stresses, particularly cold stress (Zhang et al., 2025). In *Medicago truncatula*, the interaction between MtMYB61 and MtMYB3 attenuates the repression of *MtCBF4* expression, leading to the up-regulation of *MtCAS15* and other downstream targets of MtCBF4, thereby enhancing plant freezing tolerance (Zhang et al., 2016b). Meanwhile, in rice, immediate and persistent responses to low temperature are coordinated through the complementary and sequential actions of the OsDREB1 and OsMYBS3 pathways (Su et al., 2010). Recent studies have also revealed the involvement of MYB transcription factors in apple cold acclimation. MdMYB88 and MdMYB124 contribute to cold tolerance through both CBF-dependent and CBF-independent pathways (Xie et al., 2018). Additionally, SlMYB4 and SlMYB88 can bind to the promoter of *SlGSTU43* and increase its expression. *SlGSTU43* enhances tomato cold resistance *via* the action of ALA (Zhang et al., 2024). In this study, we identified MdMYB14, an R2R3-MYB transcription factor, as a negative regulator of cold tolerance in apple (Figure 6). We further demonstrated that MdMYB14 repressed *MdCBF3* expression by directly binding to the GTGGT motif within its promoter (Figure 7). Moreover, we identified 1,785 of 4,835 putative MdMYB14 target genes involved in the cold stress response in apple (Figure 7H), indicating the important role of MdMYB14 in the cold response network.

Our results showed that MdRAD5B not only regulated *MdCBF3* expression through the MdRAD5B-MdLSD1-MdMYB14 module but also modulated chromatin accessibility at the *MdCBF3* locus. This similar regulatory paradigm has garnered increasing empirical support in plants over recent years. In *Arabidopsis*, the key transcription factor LEAFY (LFY) not only directly promotes the expression of *TERMINAL FLOWER 1* (*TFL1*) but also participates in activating *APETALA1* (*AP1*) and *CAULIFLOWER* (*CAL*), which themselves act as upstream regulators of *TFL1* expression (Goslin et al., 2017). This coordinated regulatory network involving LFY, AP1/CAL, and TFL1 ensures precise control of floral meristem identity and the transition to flowering. Furthermore, light conditions represent another layer of regulatory complexity. ELONGATED HYPOCOTYL5 (HY5) integrates light signaling to regulate anthocyanin biosynthesis through a dual mechanism. On the one hand, HY5 directly binds to the promoters of anthocyanin-biosynthetic genes and activates their expression. On the other hand, HY5 also directly binds to and activates the expression of *GOLDEN2-*

LIKE 2 (*GLK2*), which in turn directly binds to the promoters of anthocyanin-biosynthetic genes and promotes their expression (Zeng et al., 2025). This hierarchical regulatory cascade, wherein HY5 exerts both direct and indirect control *via* the GLK2 node, likely facilitates the fine-tuning and amplification of anthocyanin accumulation in response to varying light conditions. Collectively, we hypothesize that this coordinated regulation may represent a safeguard mechanism ensuring the robust activation of cold-responsive genes.

The conserved regulatory mode of MdRAD5B under both drought and cold stress suggests that it may function broadly in apple responses to diverse biotic and abiotic challenges. In general, MdRAD5B regulates the expression of drought- or cold-responsive genes through its chromatin remodeling function. However, although MdLSD1 recruits MdRAD5B to the promoter of *MdMYB14*, the transcription factor responsible for recruiting MdRAD5B to the loci of drought-responsive genes remains unidentified. Furthermore, under drought stress, MdRAD5B modulates H3K27me3 modification by promoting the degradation of MdLHP1, thereby affecting the expression of drought-responsive genes (Song et al., 2024). In contrast, under cold stress, MdRAD5B does not induce MdLHP1 degradation (Figure S20), indicating that MdRAD5B employs distinct regulatory mechanisms in response to different stresses. The molecular basis underlying this differential regulation warrants further investigation.

We further identified a natural variation in *RAD5B*, an A to G substitution at position 1,780 of the CDS ($RAD5B^{CDS1780}$), resulting in a T to A change at amino acid residue 594 of the RAD5B protein (Figure 8D; Figures S15 and S19). This variant exhibited weaker ATPase activity and a compromised ability to enhance cold tolerance (Figure 8A–C; Figures S17 and S18). We also carried out genotypic frequency analysis of the SNP at the $RAD5B^{CDS1780}$ locus, and confirmed that this SNP, which could serve as a molecular marker for breeding cold-tolerant apple varieties, was primarily subjected to artificial selection (Figure 8E and F; Figure S19). We propose two mechanistic explanations for the diminished ATPase activity. First, the T-to-A substitution may induce conformational changes that weaken ATP binding. Second, since T represents a potential phosphorylation site, its replacement with A could alter the phosphorylation status of MdRAD5B, thereby modulating its enzymatic activity.

In summary, we demonstrated that MdRAD5B modulated the expression of cold-responsive genes through its chromatin remodeling function. We also identified MdLSD1 as an interacting partner of MdRAD5B, and this interaction contributed to the repression of *MdMYB14*, a negative regulator of cold tolerance that suppresses *MdCBF3* expression. These findings not only underscore the importance of chromatin remodeling factors in plant cold adaptation but also provide valuable genetic targets for improving stress resilience in fruit crops through molecular breeding.

MATERIALS AND METHODS

Plant materials and growth conditions

The *MdRAD5B* transgenic plants were generated (Song et al., 2024) using GL-3 (a progeny of 'Royal Gala') as the transformation background (Dai et al., 2013). Tissue-cultured GL-3 and transgenic plants were cultivated on MS medium containing 4.43 g MS salts, 30 g sucrose, 0.2 mg 6-BA, 0.2 mg IAA, and 7.5 g agar per liter, with a pH of 5.8–6.0 under long-day conditions (14 h light and 10 h dark) at 25°C. Subsequently, four-week-old tissue-cultured GL-3 plants and transgenic plants were rooted on 1/2 MS medium (2.23 g MS salts, 20 g sucrose, 0.5 mg IBA, 0.5 mg IAA, and 7.5 g agar per liter, with a pH of 5.8–6.0) under long-day conditions at 25°C for 1 month (Xie et al., 2018). Then, GL-3 and *MdRAD5B* transgenic plants were transplanted into soil and grown under long-day conditions at 25°C in a growth chamber.

Apple calli (*Malus domestica* cv. Orin) were cultivated on MS medium, which contains 4.43 g MS salts, 30 g sucrose, 0.4 mg 6-BA, 1.5 mg 2,4-D, and 7.5 g agar per liter with a pH of 5.8–6.0, in darkness at 22°C as previously described (Hou et al., 2025).

Cold stress treatment

Two-month-old GL-3 and *MdRAD5B* transgenic plants were used to evaluate freezing tolerance. For cold acclimation, plants were grown under long-day conditions at 4°C for 1 week to detect cold tolerance and 2 weeks to measure the anthocyanin content, respectively.

The ion leakage assay was performed as described previously (Chen et al., 2022). Mature leaves of GL-3 and *MdRAD5B* transgenic plants were detached with or without cold acclimation and cut into leaf discs (diameter=8 mm) for measurement of conductivity.

To evaluate the survival rate of apple plants under freezing stress, we exposed whole plants to −10°C for 20 min without cold acclimation and 25 min after cold acclimation for freezing assays, then recovered them under long-day conditions at 25°C for 15 d (Shen et al., 2023).

To evaluate the cold tolerance of apple calli, we subjected 1-week-old calli to 4°C for 1 week. The calli grown at 22°C were used as control (Hou et al., 2025).

RNA extraction, RNA-seq, and qRT-PCR analysis

Total RNA extraction and qRT-PCR analysis were performed as described previously (Xie et al., 2021). Total RNA was extracted from leaves or calli using the CTAB method (Chang et al., 1993), and DNA was removed by RNase-free DNase I (Thermo Fisher Scientific, USA).

RNA-seq assay was performed as described previously (Song et al., 2024). Two-month-old GL-3 and *MdRAD5B-RNAi* transgenic plants were exposed to 4°C for 8 h, and leaves were then collected for RNA extraction.

For qRT-PCR analysis, total RNA was reverse transcribed to the first-strand cDNA using a HiScript II 1st Strand cDNA Synthesis Kit (+gDNA wiper) (Vazyme, China). qPCR was performed using Hieff Unicon Universal TaqMan multiplex qPCR master mix (Yeasen, China). Relative gene expression was calculated by the $2^{-\Delta\Delta C_t}$ method (Livak and Schmittgen, 2001), and *MdMDH* was used as the reference gene as described previously (Song et al., 2024). Primers used are listed in Table S2. Three biological replicates were used.

Generation of apple calli

Generation of apple calli was performed as described previously

(An et al., 2021). The *MdRAD5B*-OE and *MdLSD1*-anti transgenic calli were obtained previously (Song et al., 2024). The CDS of *MdMYB14* and *MdLSD1* fused with a Myc-tag were individually cloned into the pK2GW7 vector to generate over-expressing plasmids, and an antisense fragment (167 bp) of the *MdLSD1* CDS was inserted into the pH2GW7 vector to generate the knockdown plasmid. The resulting plasmids were then transformed into *Agrobacterium* strain EHA105 and used for transformation. In addition, the EHA105 strain with the pK2GW7-*MdRAD5B* plasmid was transformed into *MdLSD1*-anti calli to generate double transgenic calli. The primers used are listed in Table S2.

MNase-qPCR

MNase-qPCR was performed as described previously with some modifications (Kaster and Laubinger, 2016), and data were analyzed accordingly. Two-month-old GL-3 and *MdRAD5B* transgenic plants were exposed to 4°C for 8 h, and leaves were then collected for MNase-qPCR. Collected leaves were ground into a fine powder and nuclei were then extracted using cold 1× homogenization buffer (440 mmol/L sucrose, 1.25% Ficoll, 2.5% Dextran T40, 10 mmol/L $MgCl_2$, 20 mmol/L HEPES-KOH pH 7.4, 0.5% Triton X-100, 5 mmol/L DTT, 1 mmol/L Pefabloc, 1× protease inhibitors (Roche, Switzerland), and 167 μmol/L β-mercaptoethanol). About 50,000 nuclei were collected using density gradient centrifugation. The isolated chromatin was digested with 0.1 units/μL (final concentration) micrococcal nuclease (TaKaRa, Japan) in the reaction buffer at 37°C for 15 min. Digestions were stopped with 10 mmol/L EGTA. Mononucleosomes were then excised from agarose gels (1.5%), and DNA samples were extracted as described (Kaster and Laubinger, 2016) and used for MNase-qPCR. Nucleosome occupancy analysis by qPCR requires the design of tiled oligonucleotides spanning the region of interest. Each amplicon should be approximately 100 bp in length, with a 20 bp overlap between adjacent amplicons. Here, we used the DNA sequences 1,000 bp upstream and 300 bp downstream of the TSS of the locus for oligonucleotide design. The primers used are listed in Table S2. Three biological replicates were used.

Dual-LUC assay

Dual-LUC assays were performed as previously described (Niu et al., 2022). $MdMYB14_{pro}$:LUC, $MdJAZ6_{pro}$:LUC, and $MdCBF3_{pro}$:LUC were individually introduced into the pGreen0800-LUC vector as reporter genes, and *35S:MdLSD1*, *35S:MdMdRAD5B*, and *35S:MdMYB14* were cloned into the pGreen62-SK vector as effectors. The resulting plasmids were then transformed into *Agrobacterium* strain GV3101, and the different combinations were injected into *Nicotiana tabacum* leaves. The firefly and Renilla luciferase signals were measured using the Dual-LUC Reporter Assay Kit (Yeasen) after infiltration for 3 d. Experiments were repeated 3 times with at least 8 biological replicates.

ChIP-qPCR assay

ChIP-qPCR assays were performed as previously described (Xie et al., 2018). Briefly, the *MdMYB14*-OE or *MdLSD1*-OE transgenic calli labeled with a Myc-tag were cross-linked in 1% (w/v) formaldehyde under vacuum for 15 min. Nuclei were isolated and sonicated into DNA fragments around 0.5 kb. The sonicated samples were then incubated with α-Myc antibody (ABclonal, China) overnight with rotation at 4°C. The associated DNA separated from the beads was used for ChIP-qPCR. The primers for ChIP-qPCR are listed in Table S2. Three biological replicates were used.

EMSA

EMSA assays were performed as described previously (Xie et al., 2018). The CDS of *MdLSD1* and *MdMYB14* were cloned into the pGEX-4T-1 vector. The fusion proteins GST-MdLSD1 and GST-MdMYB14 were induced with 0.8 mmol/L IPTG in *E. coli* at 28°C, shaken at 120 r/min for 8 h, and purified using Pierce Glutathione Agarose Resin (Thermo Fisher Scientific). The probes were synthesized by Sangon Biotech Co., Ltd. (Shanghai, China). EMSA was then performed using a LightShift Chemiluminescent EMSA Kit (Thermo Fisher Scientific).

Y2H assay

Y2H assays were carried out according to the manufacturer's manual (Clontech, USA). Full-length and truncated *MdRAD5B* were individually cloned into pGBKT7 (Song et al., 2024), and the CDS of *MdLSD1* was cloned into pGADT7. To detect the self-activation of MdLSD1, we cloned the CDS of *MdLSD1* into pGBKT7. pGBKT7-53 and pGADT7-T were used as positive controls (P53), and pGBKT7-Lam and pGADT7-T were used as negative controls (Lam). The SD/-Leu-Trp medium was used to select positive clones, and the SD/-Leu-Trp-His-Ade medium was used for further selection.

Split-luciferase assay

Split-luciferase assays were performed as described previously (Chen et al., 2022). The CDS of *MdRAD5B* was cloned into pCAMBIA1300-cLUC (Song et al., 2024), and the CDS of *MdLSD1* was cloned into pCAMBIA1300-nLUC. The resulting plasmids and empty vector plasmids were then transformed into *Agrobacterium* strain GV3101, and the different combinations were injected into *Nicotiana tabacum* leaves. Luciferase expression signal was detected after infiltration for 3 d using Lumazone Pylon 2048B (Princeton, USA).

Western blot and Co-IP assay

Western blot assays were performed as described previously (Chen et al., 2022). The CDS of *MdRAD5B* was cloned into a modified pCAMBIA1300 vector, which contains a Flag-tag (Song et al., 2024), and the CDS of *MdLSD1* was cloned into pEarleyGate203, which contains a Myc-tag. The resulting plasmids were then transformed into *Agrobacterium* strain GV3101, and the different combinations were injected into *Nicotiana tabacum* leaves. Total proteins from *Nicotiana tabacum*, GL-3, and *MdRAD5B* transgenic plants were extracted with a buffer containing 2 mmol/L EDTA (pH 8.0), 50 mmol/L Tris-HCl (pH 7.5), 150 mmol/L NaCl, and 0.1% NP-40. After separation on a 10% SDS polyacrylamide gel, the proteins were detected by the α-GST (ABclonal), α-Myc (ABclonal), or α-Flag (ABclonal) antibody.

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Co-IP assay was performed as previously described (Chen et al., 2022). Briefly, total proteins were extracted and incubated with the α-Flag antibody at 4°C for 8 h. The protein A/G agarose beads (Thermo Fisher Scientific) were used to collect the immunocomplexes and then washed with immunoprecipitation buffer (50 mmol/L Tris-HCl, pH 8.0, 150 mmol/L NaCl, 2 mmol/L EDTA, 1 mmol/L DTT, 10% glycerol, 0.15% Triton X-100, 1 mmol/L PMSF, and 1× Halt protease inhibitor cocktail (Thermo Fisher Scientific)). The immunocomplexes with beads were resuspended in 1× SDS-PAGE loading buffer. After separation on a 10% SDS polyacrylamide gel, the eluted proteins were detected by α-Myc antibody.

ATPase activity assay

The ATPase activities of MdRAD5B and MdRAD5B^{T594A} were detected as previously described (Guan et al., 2013). First, the CDS of *MdRAD5B* was cloned into the pGEX-4T-1 vector. The pGEX-4T-1-*MdRAD5B* vector was used as a template to generate a mutant form, pGEX-4T-1-*MdRAD5B*^{T594A}, in which the threonine at the 594th position was replaced by alanine. The fusion proteins GST-MdRAD5B and GST-MdRAD5B^{T594A} were induced with 0.8 mmol/L IPTG in *E. coli* at 28°C, shaken at 120 r/min for 8 h, and purified using Pierce Glutathione Agarose Resin (Thermo Fisher Scientific). The ATPase assays were carried out as reported previously (Guan et al., 2013), using 50 nmol/L GST-MdRAD5B or GST-MdRAD5B^{T594A}. Three biological replicates were used.

Statistical tests

Student's *t*-test was performed and statistically significant differences were indicated by *, $P<0.05$.

Data availability

RNA-seq data were submitted to the Sequence Read Archive of the NCBI under the accession number PRJNA898595. Sequence data from this article can be found in the GDR databases under the following accession numbers: MdRAD5B (MD16G1213200), MdCBF3 (MD07G1262900), MdCCA1 (MD01G1090900), MdLEA4-5 (MD12G1057800), MdMYB12 (MD03G1297100), MdMYB14 (MD05G1197600), MdF3H (MD02G1132200), MdF3'H (MD06G1201700), MdCHI (MD07G1233400), MdANS (MD06G1071600), MdLAR (MD16G1048500), MdUFGT (MD01G1234400), MdJAZ6 (MD09G1178600), and MdLSD1 (MD05G1263500).

Compliance and ethics

The authors declare that they have no conflict of interest.

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Supporting information

The supporting information is available online at https://doi.org/10.1007/s11427-025-3445-5. The supporting materials are published as submitted, without typesetting or editing. The responsibility for scientific accuracy and content remains entirely with the authors.

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